14. For the given data find the value of $\Delta^3 y_0$ is

x	5	6	9	11
y	12	13	15	18

(a) 1 (b) 0 (c) 2 (d) –1

Miscellaneous Problems

1. If $f(x)=e^{ax}$ then show that $f(0)$, $\Delta f(0), \Delta^2 f(0)$ are in G.P.

2. Prove that $i)(1+\Delta)(1-\nabla)=1$
$ii)\Delta\nabla=\Delta-\nabla$ (iii) $E\nabla=\Delta=\nabla E$

3. A second degree polynomial passes though the point (1,-1) (2,-1) (3,1) (4,5). Find the polynomial.

4. Find the missing figures in the following table:

x	0	5	10	15	20	25
y	7	11	-	18	-	32

5. Find $f(0.5)$ if $f(-1)=202$, $f(0)=175$, $f(1)=82$ and $f(2)=55$.

6. From the following data find y at x = 43 and x = 84.

x	40	50	60	70	80	90
y	184	204	226	250	276	304

7. The area A of circle of diameter 'd' is given for the following values

D	80	85	90	95	100
A	5026	5674	6362	7088	7854

Find the approximate values for the areas of circles of diameter 82 and 91 respectively.

8. If $u_0=560$, $u_1=556$, $u_2=520$, $u_4=385$, show that $u_3=465$.

9. From the following table obtain a polynomial of degree y in x.

x	1	2	3	4	5
y	1	–1	1	–1	1

10. Using Lagrange's interpolation formula find a polynomial which passes through the points (0, –12), (1, 0), (3, 6) and (4,12).

Summary

In this chapter we have acquired the knowledge of

- $\Delta f(x)=f(x+h)-f(x)$
- $\nabla f(x)=f(x)-f(x-h)$
- $\nabla f(x+h)=\Delta f(x)$
- $Ef(x)=f(x+h)$
- $E^n f(x)=f(x+nh)$
- Newton's forward interpolation formula:

$$y_{(x=x_0+nh)}=y_0+\frac{n}{1!}\Delta y_0+\frac{n(n-1)}{2!}\Delta^2 y_0+\frac{n(n-1)(n-2)}{3!}\Delta^3 y_0+\ldots$$

- Newton's backward interpolation formula:

$$y_{(x=x_n+nh)}=y_n+\frac{n}{1!}\nabla y_n+\frac{n(n+1)}{2!}\nabla^2 y_n+\frac{n(n+1)(n+2)}{3!}\nabla^3 y_n+\ldots$$

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